

Recap.

① Expected value: $E(X) = \begin{cases} \sum_{x \in \Omega} x p_X(x) \\ \int_{-\infty}^{\infty} x p_X(x) dx \end{cases}$ "Average value of a RV."

Variance: $Var(X) = E((X - E(X))^2)$ "Fluctuation of a RV."
 $= E(X^2) - (E(X))^2$

Covariance: $cov(X, Y) = E((X - E(X))(Y - E(Y)))$ "How deep are RVs related."
 $= E(XY) - E(X)E(Y)$

② $E(aX + bY) = aE(X) + bE(Y)$

$Var(aX + b) = a^2 Var(X)$

$cov(X, Y) = cov(Y, X)$, $cov(aX + bY, Z) = a cov(X, Z) + b cov(Y, Z)$

If X is independent of Y , $X \perp Y$ $\left\{ \begin{array}{l} E(XY) = E(X)E(Y) \\ Var(X+Y) = Var(X) + Var(Y) \\ cov(X, Y) = 0 \text{ (converse statement false)} \end{array} \right.$

③ Independence: "The distribution of X doesn't effect the distribution of Y ."

If $X \perp Y$, then $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$, $F_{X,Y}(x, y) = F_X(x) \cdot F_Y(y)$.

Uncorrelatedness: " X, Y is not linear-related, weaker than $X \perp Y$."

$cov(X, Y) = 0$.

Exercises.

Ex 1.

① $Var(aX + bY) = E((aX + bY)^2) - (E(aX + bY))^2$
 $= E(a^2 X^2 + 2abXY + b^2 Y^2) - (aE(X) + bE(Y))^2$
 $= a^2 E(X^2) + 2ab E(XY) + b^2 E(Y^2) - a^2 E(X)^2 - 2ab E(X)E(Y) - b^2 E(Y)^2$
 $= a^2 (E(X^2) - E(X)^2) + b^2 (E(Y^2) - E(Y)^2) + 2ab (E(XY) - E(X)E(Y))$
 $= a^2 Var(X) + b^2 Var(Y) + 2ab cov(X, Y)$

② $cov(aZ_1 + bZ_2, Z_3) = E[(aZ_1 + bZ_2 - E(aZ_1 + bZ_2))^T (Z_3 - E(Z_3))]$
 $= E[(aZ_1 - aE(Z_1) + bZ_2 - bE(Z_2))^T (Z_3 - E(Z_3))]$
 $= E[a(Z_1 - E(Z_1))^T (Z_3 - E(Z_3)) + b(Z_2 - E(Z_2))^T (Z_3 - E(Z_3))]$
 $= a cov(Z_1, Z_3) + b cov(Z_2, Z_3)$

Ex 2. Hint: Taylor series: $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

$$\begin{aligned} \text{(i)} \quad E(X) &= \sum_{n=0}^{\infty} n \frac{\lambda^n}{n!} e^{-\lambda} \\ &= 0 + \sum_{n=1}^{\infty} n \frac{\lambda^n}{n!} e^{-\lambda} \\ &= \sum_{n=1}^{\infty} \frac{\lambda^n}{(n-1)!} e^{-\lambda} \\ &= \lambda e^{-\lambda} \sum_{n=1}^{\infty} \frac{\lambda^{n-1}}{(n-1)!} \quad (\text{define } m=n-1) \\ &= \lambda e^{-\lambda} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} \\ &= \lambda. \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= E(X^2) - E(X) + E(X) - (E(X))^2 \\ &= E(X^2 - X) + E(X) - (E(X))^2 \\ &= \lambda - \lambda^2 + \sum_{n=0}^{\infty} n(n-1) \frac{\lambda^n}{n!} e^{-\lambda} \\ &= \lambda - \lambda^2 + \sum_{n=2}^{\infty} \frac{\lambda^n}{(n-2)!} e^{-\lambda} \quad (m=n-2) \\ &= \lambda - \lambda^2 + \lambda^2 e^{-\lambda} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} \\ &= \lambda. \end{aligned}$$

(ii) Define $Y: \Omega \rightarrow \{0,1\}$ with $P(Y=1)=p$, $P(Y=0)=1-p$, and $X = \sum_{i=1}^n Y_i$.
 $E(Y) = 1 \cdot P(Y=1) + 0 \cdot P(Y=0) = p$, $\text{Var}(Y) = E(Y) - (E(Y))^2 = p(1-p)$.
 $E(X) = \sum_{i=1}^n E(Y_i) = n \cdot p$, $\text{Var}(X) = \sum_{i=1}^n \text{Var}(Y_i) = np(1-p)$.

Ex 3. Hint: pdf of uniform distributed RV on domain of $[a, b]$ is $\frac{1}{b-a}$.

$$\begin{aligned} \text{(i)} \quad X \perp Y, \quad p_{X,Y}(x,y) &= p_X(x) \cdot p_Y(y). \\ E(XY) &= \iint_{\mathbb{R}^2} xy p_{X,Y}(x,y) dx dy \\ &= \int_{\mathbb{R}} x p_X(x) dx \int_{\mathbb{R}} y p_Y(y) dy \\ &= E(X) E(Y). \end{aligned}$$

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = 0.$$

(ii) Define $X \sim \text{unif}(-1, 1)$, $f_X(x) = \begin{cases} \frac{1}{2} & \text{if } -1 \leq x \leq 1, \\ 0 & \text{otherwise} \end{cases}$, and $Y = X^2$.

① Uncorrelatedness:

$$E(X) = \frac{(-1+1)}{2} = 0, \quad E(XY) = E(X^3) = \int_{-1}^1 x^3 \cdot \frac{1}{2} dx = 0.$$

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = 0.$$

② Non-independence:

$$F_X(x) = \int_{-1}^x p_X(t) dt = \frac{x+1}{2} \quad (-1 \leq x \leq 1).$$

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \sqrt{y} \quad (0 \leq y \leq 1).$$

$$\begin{aligned} F_{X,Y}(x,y) &= P(X \leq x, X^2 \leq y) \\ &= P(X \leq x, -\sqrt{y} \leq X \leq \sqrt{y}) \\ &= P(-\sqrt{y} \leq X \leq \min(\sqrt{y}, x)) \\ &= \frac{1}{2} (\min\{x, \sqrt{y}\} + \sqrt{y}) \neq F_X(x) \cdot F_Y(y). \end{aligned}$$